
SaaS Lead Generation & Revenue Intelligence Pipeline

BILL SWIFT

An end-to-end automation project for Bill Swift, a usage-based billing SaaS company

A portfolio project by Praveen Sambasivam

What is Bill Swift?

Bill Swift is a fictional SaaS company built as the business scenario for this portfolio project — modeled on Chargebee, a real subscription billing and revenue management platform used by SaaS companies. Bill Swift gives every part of this pipeline a reason to exist: a webinar, a scored lead, a CRM record, and a revenue forecast that a real billing-SaaS company's sales and finance teams would actually use.

What is this project about?

This project is an automated, end-to-end SaaS lead-generation and revenue-intelligence pipeline. From the moment someone registers for a webinar to the moment their deal closes, every step — scoring, CRM sync, email nurture, revenue prediction, and reporting — runs without manual intervention.

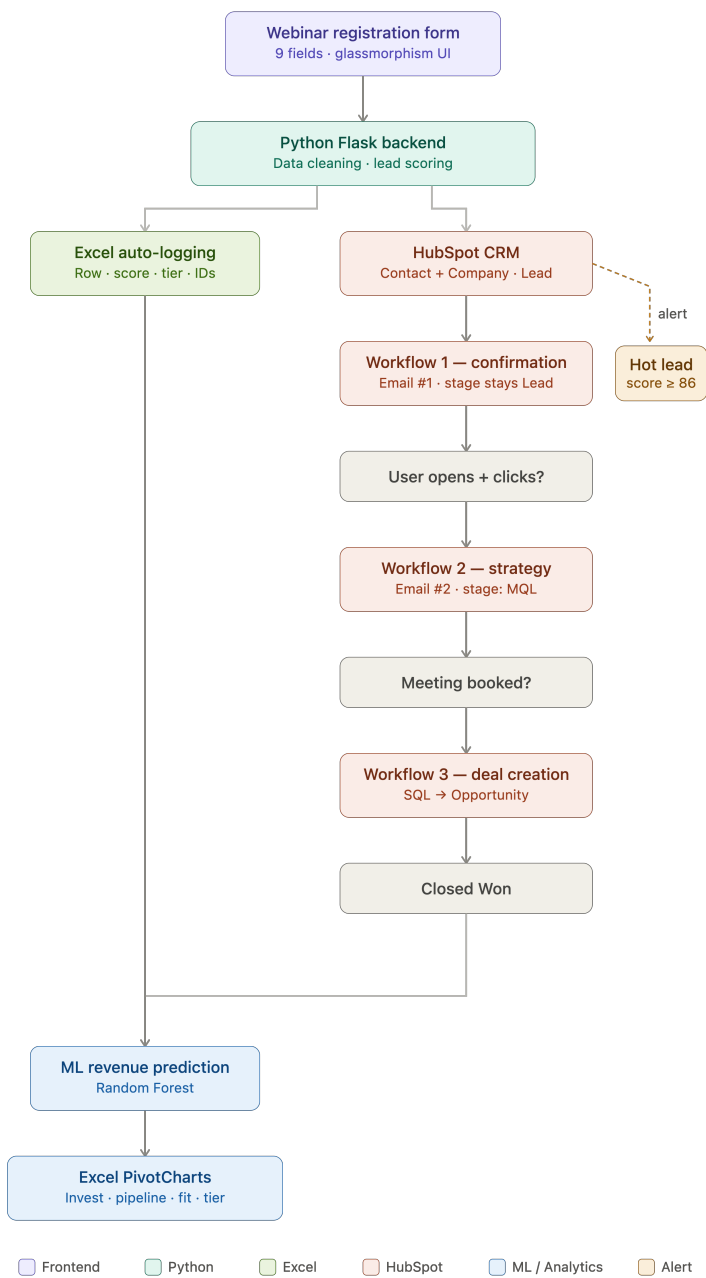
What problem does this project solve?

Every SaaS company faces the same gap between a new lead and a closed deal — leads come in faster than anyone can manually qualify, track, or follow up with. For Bill Swift, I built a system to close that entire gap automatically:

- **Lead qualification** — a 6-signal, 100-point scoring engine scores every lead automatically based on the details captured at signup (company size, transaction volume, current billing setup, and more), so sales isn't guessing who to prioritize.
- **CRM and data consistency** — every lead syncs to Excel and HubSpot simultaneously with matching IDs, keeping one reliable source of truth instead of two drifting records.
- **Hot lead response time** — an automated alert flags any lead scoring above 85, so high-value prospects aren't missed while sitting in a queue.
- **Structured follow-up** — tiered nurture workflows move leads through Lead → MQL → SQL → Opportunity → Closed Won without manual handoffs.
- **Revenue visibility** — an ML revenue-prediction model feeds CXO-ready PivotCharts, showing leadership where the pipeline is strongest and where to invest.

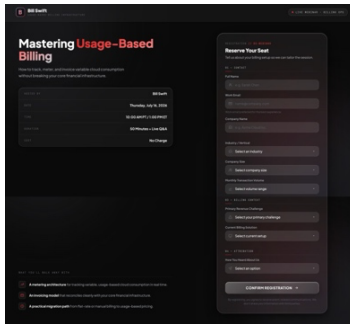
Pipeline Overview

Here's how a lead moves through the system, end to end:

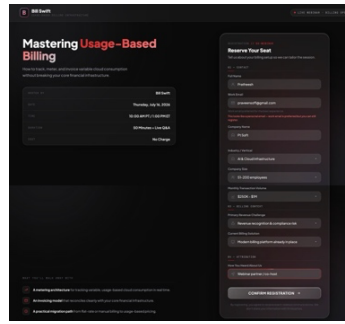


1. Webinar registration form

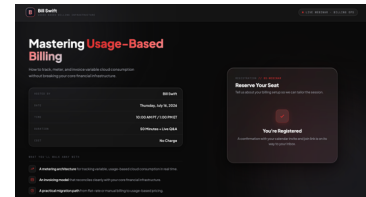
Lead-gen entry point — a webinar signup form that captures contact, company, and billing context in one step.



Full webpage — event details and registration form



Form mid-fill — personal email soft warning shown



Success screen — confirms the pipeline is live

2. Python Flask backend

Receives the form data, cleans it, and triggers scoring — all in one request.

3. Lead scoring engine

Scores each lead out of 100 across 6 signals; sorts into Hot / Warm / Cold.

Signal	Max
Monthly Transaction Volume	28
Primary Revenue Challenge	24
Current Billing Solution	19
Company Size	14
How You Heard About Us	10
Email Type	5

Tiers: 0-35 Cold · 36-65 Warm · 66-100 Hot

Six weighted scoring signals feed a 100-point scale, split into three tiers.

4. Excel auto-logging

Auto-appends every lead to Excel with score, tier, and IDs — no manual entry.

Timestamp	Name	Email	Company	Company	Industry	Industry	Company	Monthly	Primary	Current	Heard From	Lead Score	Lead Tier	HubSpot	HubSpot	Predict
2026-07-01	Praveen Sai	praveen.dur	Prvn Tech	Personal Er	edtech		201-1000	250k-1m	manual-in	stripe-payp	google	58	Warm	51244505	53152383	152837.3
2026-07-01	Sarah Chen	sarah@veri	Vertex AI	vertexai.io	ecommerce		201-1000	50k-250k	usage-mete	in-house	newsletter	65	Warm	51244541	53157054	227258
2026-07-01	David Ross	dross@saa	SaaSflow	saasflow.cc	proptech-hot		201-1000	250k-1m	usage-mete	in-house	partner	75	Hot	51244342	53153833	271457.6
2026-07-01	Elena Rostk	elena@cyb	Cybercore	cybercore.ai	fintech		11-50	10m+	research	stripe-payp	google	60	Warm	51245502	53149321	184046.1
2026-07-01	Elena Rostk	elena@cyb	Cybercore	cybercore.ai	ai-cloud		11-50	250k-1m	research	legacy-ente	referral	45	Warm	51245502	53149321	126484.1
2026-07-01	Marcus Var	marcus@p	Paybolt	paybolt.fin	ecommerce		51-200	1m-10m	research	legacy-ente	newsletter	51	Warm	51238017	53154755	157615.1
2026-07-01	Chloe Tana	c.tanaka@	Edulearn	edulearn.ci	edtech		1-10	under-50k	rev-recogni	in-house	partner	47	Warm	51235872	53147174	180323.1
2026-07-01	Liam O'Con	liam@stres	Streamvib	streamvibe	digital-media		1-10	250k-1m	usage-mete	none-man	other	66	Hot	51242147	53154585	238010.9
2026-07-01	Amara Okai	a.okafor@	Nexthome	nexthome.	proptech-hot		51-200	10m+	research	stripe-payp	linkedin	65	Warm	51235194	53153292	193846.6
2026-07-01	Jacob Miller	j.miller@bi	Biotrack	biotrack.or	ai-cloud		1000+	50k-250k	manual-in	none-man	google	62	Warm	51244507	53149501	171892.2
2026-07-01	Pratish	praveensof	Pt Soft	Personal Er	ai-cloud		51-200	250k-1m	rev-recogni	modern-pli	partner	49	Warm	51238751	53152383	189583.2

Every submission appends a new row — score, tier, and HubSpot IDs included.

Creates Contact + Company in HubSpot; handles duplicate conflicts automatically.

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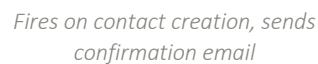
HubSpot Contacts — auto-created on every submission

HubSpot Companies — linked to matching contacts

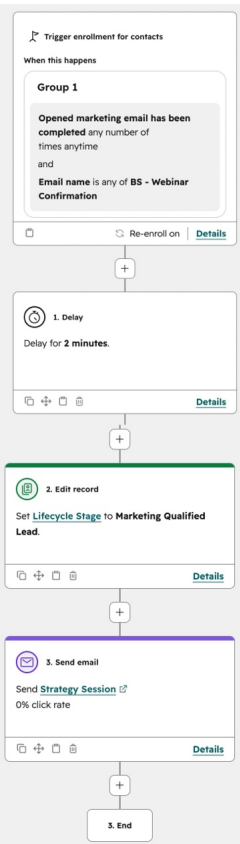
4 workflows: confirmation email → MQL upgrade → SQL/deal creation → hot lead alert.

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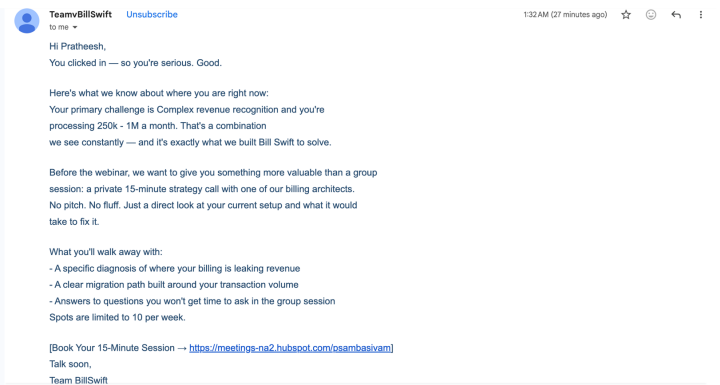
Workflow 1 — registration confirmation



Workflow 2 — MQL upgrade & strategy session

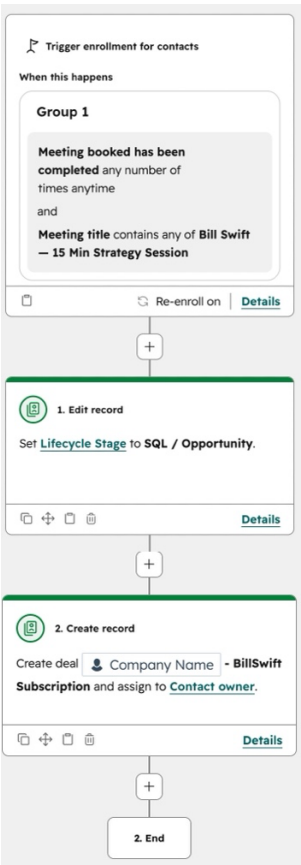


Email open triggers MQL upgrade,
sends Email #2



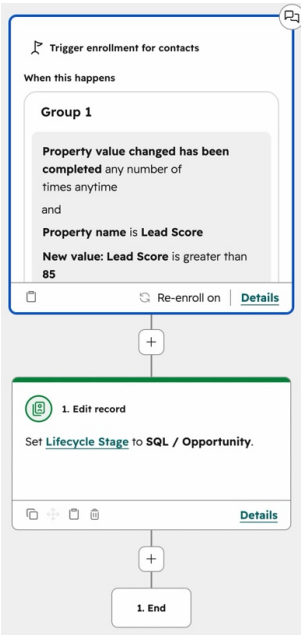
Email #2 — dynamic tokens populate real challenge + volume

Workflow 3 — meeting booked → SQL & deal creation



Meeting booked triggers SQL/Opportunity and auto-creates a deal.

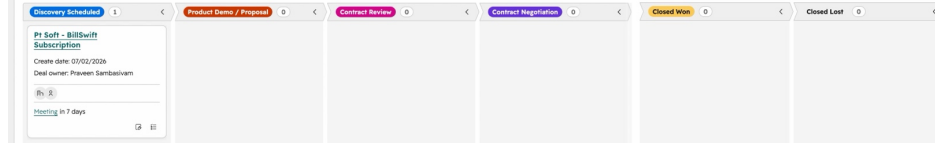
Workflow 4 — hot lead alert (Lead Score > 85)



Lead Score above 85 triggers an instant SQL upgrade and sales alert — independent of the email sequence.

7. Deal creation & Closed Won

Meeting booked → deal auto-created in pipeline → progresses to Closed Won.



Deal auto-created at Discovery Scheduled — no manual entry into the pipeline.

8. ML revenue prediction

Random Forest model predicts contract value; written back to Excel.

Timestamp	Name	Email	Company Name	Company Domain	Industry	Industry Size	Company Size	Monthly Revenue Range	Primary Revenue Source	Current Billing Interval	Lead Score	Lead Type	Lead Tier	Model Score (0-1)	Model Score (0-1)	Actual Revenue
2020-01-01 00:00:00 UTC	John Doe	john.doe@company.com	Company A	companya.com	Technology	Small	10-50	\$100-\$500	Software	Monthly	58	Warm	58	0.58	0.58	\$15,000.00
2020-01-01 00:00:00 UTC	Jane Smith	jane.smith@company.com	Company B	companyb.com	Marketing	Medium	50-100	\$500-\$1000	Marketing	Quarterly	65	Warm	65	0.65	0.65	\$22,725.8
2020-01-01 00:00:00 UTC	John Doe	john.doe@company.com	Company C	companyc.com	Technology	Small	10-50	\$100-\$500	Software	Monthly	75	Hot	75	0.75	0.75	\$27,145.7
2020-01-01 00:00:00 UTC	Jane Smith	jane.smith@company.com	Company D	companyd.com	Marketing	Medium	50-100	\$500-\$1000	Marketing	Quarterly	60	Warm	60	0.60	0.60	\$18,404.6
2020-01-01 00:00:00 UTC	John Doe	john.doe@company.com	Company E	companye.com	Technology	Small	10-50	\$100-\$500	Software	Monthly	45	Warm	45	0.45	0.45	\$12,648.4
2020-01-01 00:00:00 UTC	Jane Smith	jane.smith@company.com	Company F	companyf.com	Marketing	Medium	50-100	\$500-\$1000	Marketing	Quarterly	51	Warm	51	0.51	0.51	\$15,761.5
2020-01-01 00:00:00 UTC	John Doe	john.doe@company.com	Company G	companyg.com	Technology	Small	10-50	\$100-\$500	Software	Monthly	47	Warm	47	0.47	0.47	\$18,032.3
2020-01-01 00:00:00 UTC	Jane Smith	jane.smith@company.com	Company H	companyh.com	Marketing	Medium	50-100	\$500-\$1000	Marketing	Quarterly	66	Hot	66	0.66	0.66	\$23,801.0
2020-01-01 00:00:00 UTC	John Doe	john.doe@company.com	Company I	companyi.com	Technology	Small	10-50	\$100-\$500	Software	Monthly	65	Warm	65	0.65	0.65	\$19,384.6
2020-01-01 00:00:00 UTC	Jane Smith	jane.smith@company.com	Company J	companyj.com	Marketing	Medium	50-100	\$500-\$1000	Marketing	Quarterly	62	Warm	62	0.62	0.62	\$17,189.2
2020-01-01 00:00:00 UTC	John Doe	john.doe@company.com	Company K	companyk.com	Technology	Small	10-50	\$100-\$500	Software	Monthly	49	Warm	49	0.49	0.49	\$18,958.3

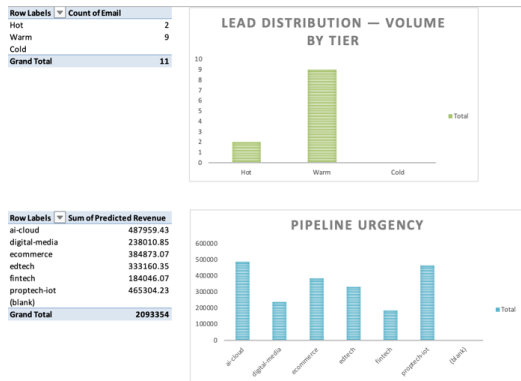
200-row historical dataset used to train the Random Forest model at server startup.

Lead Score	Lead Tier	HubSpot	HubSpot	Predict
58	Warm	51244505	33152383	152837.3
65	Warm	51248541	33157054	22725.8
75	Hot	51244342	33153833	271457.6
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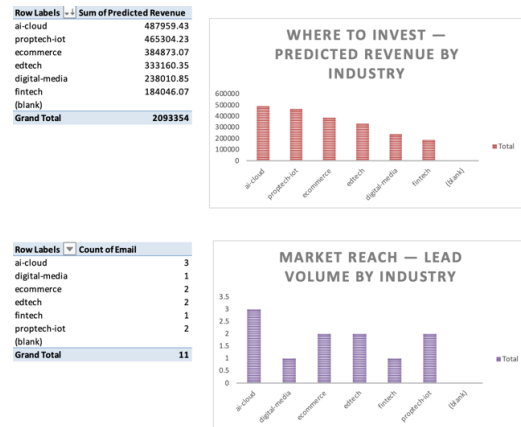
Predicted revenue is written back into the same Excel row as the lead score.

9. Excel PivotCharts

4 live charts: revenue by industry, lead volume, tier distribution, pipeline urgency.



Lead Distribution by Tier (2 Hot · 9 Warm) and Pipeline Urgency



Where to Invest — revenue by industry, and Market Reach by volume

Project Summary

- Zero manual steps — from form submission to CRM contact, ML prediction, and email delivery, everything is automated.
- Server-side lead scoring — scores are calculated in Python and cannot be tampered with from the browser.
- Dynamic email personalization — Email #2 injects real CRM properties into each send.
- Full audit trail — every row contains a timestamp, cleaned data, lead score, tier, HubSpot IDs, and predicted revenue.